

No. of Functional Features Included in the Solution

Date	1 July 2026
Team ID	XXXXXX
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Data Collection	Collect historical rainfall, weather and flood-related datasets	Dataset Module	Done	Provides training data for model
2	Data Preprocessing	Clean missing values and prepare dataset for training	Preprocessing Module	Done	Improves data quality and accuracy
3	Feature Selection	Select important parameters like rainfall, humidity, water level	ML Processing Module	Done	Enhances prediction performance
4	Model Training	Train machine learning model using flood dataset	Training Module	Done	Builds prediction engine
5	Flood Prediction	Predict flood occurrence based on input parameters	Prediction Module	Done	Enables early flood warning
6	Accuracy Evaluation	Measure model performance and compare prediction accuracy	Testing Module	Done	Validates system reliability
7	Result Visualization	Display graphs and prediction outputs	Visualization Module	Done	Helps interpret prediction results

Feature Summary:

Metric	Count / Value
Total Features Planned	7

Total Features Implemented	7
Core / Must-Have Features	5
Additional / Nice-to-Have Features	Real-time weather API integration and automated flood alert notification system for early disaster warning.
Features Tested & Verified	All implemented modules were tested successfully including dataset loading, preprocessing, machine learning model training, prediction output generation, and result visualization.

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	1	Prediction Output Dashboard
2	Backend / Logic	3	Data Preprocessing, Model Training, Flood Prediction
3	Database / Storage	1	Dataset Storage (CSV Files)
4	API / Integration	1	Weather Data Integration (Future Scope)
5	Security / Authentication	1	Secure Access for Admin Monitoring